MTH105

Timed Online Assignment – July Semester 2021

Fundamentals of Mathematics

Monday, 15 November 2021 **4:00 pm – 6:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **EIGHT (8)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

(a) Let p and q be statements. Employ a truth table to verify that $\sim p \leftrightarrow q$ is logically equivalent to $(\sim p \rightarrow q) \wedge (q \rightarrow \sim p)$.

(6 marks)

(b) Construct a chain of logical connectives to show that $\sim p \leftrightarrow q$ is logically equivalent to $p \leftrightarrow \sim q$. Do not use truth tables here and give a reason for each line.

(5 marks)

Question 2

(a) Consider the quantified statement $\exists x \forall y (\sim B(x) \wedge C(y))$.
Write down and simplify the negation of the statement, giving your final answer with no negation symbol.

(3 marks)

(b) Consider the following argument:
All tigers are ferocious.
Some tigers do not eat chocolate.
Therefore, some ferocious animals do not eat chocolate.

Let $T(x)$, $F(x)$ and $C(x)$ be the predicates "x is a tiger" and "x is ferocious" and " x eats chocolate" respectively.

(i) Rewrite the above argument using the predicates $T(x)$, $F(x)$ and $C(x)$ together with the universal and/or existential quantifier.

(3 marks)

(ii) Use the rules of inference to prove the validity of the argument using the above predicates and/or universal and existential quantifier. Give a reason for each line of your proof.

(6 marks)

Question 3

(a) Consider the following statement:

If 3 divides the integer n^2, then 3 divides the integer n.

(i) State the contrapositive of the above statement.

(2 marks)

(ii) Use the method of proof by contrapositive to prove the original statement.

(6 marks)

(b) Prove that there are no pairs of integers (x, y) which satisfies $x^2 + y^2 = 7007$ or give a counter-example which shows otherwise.

(7 marks)

Question 4

A sequence is defined by $u_0 = 1$, $u_1 = 2$ and $u_2 = 3$ and

$$u_n = 7u_{n-1} - 15u_{n-2} + 9u_{n-3}, n \geq 4.$$

Use mathematical induction to show that

$$u_n = (3 - n)3^{n-1},$$

for $n \geq 0$.

(9 marks)

Question 5

(a) Show that for any sets A, B and C,

$$(A \cup B) \cap (A \cap B)^c = (A \setminus B) \cup (B \setminus A).$$

Give the reason for each line of your proof.

(6 marks)

(b) Implement the Euclidean Algorithm to determine d = gcd(876543210, 9999) and express d as an integral linear combination of 876543210 and 9999.

(10 marks)

Question 6

(a) Let $f: \mathbb{R} \longrightarrow \mathbb{R}$ be the function defined by the rule $f(x) = x^3 - 4x$ where $\mathbb{R}$ denotes the set of real numbers.

Determine whether the function f is injective and surjective or not. Justify your answers.

(5 marks)

(b) Consider a relation R on the set of rational numbers Q. For $a, b \in Q$, we define aRb if and only if $a - b \in Z$.

(i) Show that R is reflexive, symmetric and transitive.

(4 marks)

(ii) Describe in set theoretic notation the equivalence class of $\frac{2}{3}$.

(2 marks)

Question 7

(a) The function f with domain $\mathbb{R}$ is defined by $f(x) = x^2 - 6bx$ where b is a real constant. Find in terms of b, $ff(b)$ and the range of $f(x)$.

(4 marks)

(b) The function f has an inverse if its domain is restricted to $x \le k$.
Find the largest value of k and also the inverse function $f^{-1}(x)$ in the simplest form, both in terms of b.

(6 marks)

Question 8

(a) SUSS is sending a team of 6 students selected from 6 basketball players, 6 rugby players and 5 soccer players. No student plays more than one game. The team is to consist of at least one, and not more than 3 players drawn from each sport. Giving full details of your working, find the number of ways in which the team can be selected.

(6 marks)

(b) In a sales campaign, a company gives each customer who buys its magazine a picture card. There are 10 different pictures and any customer who collects a complete set of all 10 picture cards gets a free gift. On any occasion when a customer buys a magazine, the card received is equally likely to be any of the 10 picture cards.

(i) Find the probability that the first 4 cards the customer receives all carry different pictures.

(2 marks)

(ii) Find the probability that the first 4 cards received result in the customer having exactly 3 different pictures.

(2 marks)

(iii) Two of the 10 pictures are X and Y. Find the probability that the first 4 cards received result in the customer having a picture of X or of Y (or both).

(2 marks)

(iv) At a certain stage the customer has collected 9 of the 10 pictures. Find the least value of n such that the probability of at most n more cards are needed to complete the set is at least 0.99.

(4 marks)

----- END OF PAPER -----